Structures

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Programming Fundamentals

What is a Structure?

- Structure is a collection of variables of different data types under a single name.
- For example: You want to store some information about a person: his/her name, citizenship number and salary. You can easily create different variables name, citNo, salary to store these information separately.

C++ Structures:

- If, in the future, you would want to store information about multiple persons, what you will do ?

- you'd need to create different variables for each information per person: name1, citNo1, salary1, name2, citNo2, salary2
- You can easily visualize how big and messy the code would look. Also, since no relation between the variables (information) would exist, it's going to be a scary task.

C++ Structures:

- A better approach will be to have a collection of all related information under a single name Person, and use it for every person. Now, the code looks much cleaner, readable and efficient as well.
- This collection of all related information under a single name Person is a structure.

How to declare a structure in C++ programming?

- The struct keyword defines a structure type followed by an identifier (name of the structure).
- Then inside the curly braces, you can declare one or more members (declare variables inside curly braces) of that structure. For example:
- Here a structure person is defined which has three members: name, age and salary.
- When a structure is created, no memory is allocated.
- You can imagine it as a datatype.

How to define a structure variable?

- Once you declare a structure person as above. You can define a structure variable as:
- Person ahmed;
- Here, a structure variable ahmed is defined which is of type structure Person.
- When structure variable is defined, only then the required memory is allocated by the compiler.
- the memory of float is 4 bytes, memory of int is 4 bytes and memory of string is 8 byte. Hence, 16 bytes of memory is allocated for structure variable ahmed.

How to access members of a structure?

- The members of structure variable is accessed using a dot (.) operator.
- Suppose, you want to access age of structure variable ahmed and assign it 22 to it. You can perform this task by using following code below:
- ahmed.age=22;

Example: C++ Structure

- Named Structures:

By giving a name to the structure, you can treat it as a data type. This means that you can create variables with this structure anywhere in the program at any time: like example in this slide

ructure.cpp

```cpp
#include <iostream>
using namespace std;

struct Person
{
    string name;
    int age;
    float salary;
};
int main()
{
    Person p1;
    p1.name = "Ahmed";
    p1.age = 22;
    p1.salary = 50000.50;
    cout << "Name: " << p1.name << endl;
    cout <<"Age: " << p1.age << endl;
    cout << "Salary: " << p1.salary;
    return 0;
}
```

```
C:\Users\LENOVO\Documents
Name: Ahmed
Age: 22
Salary: 50000.5
--------------------------------
Process exited after 0.09139 second
Press any key to continue . . .
```

Example: C++ Structure

•unnamed Structures:

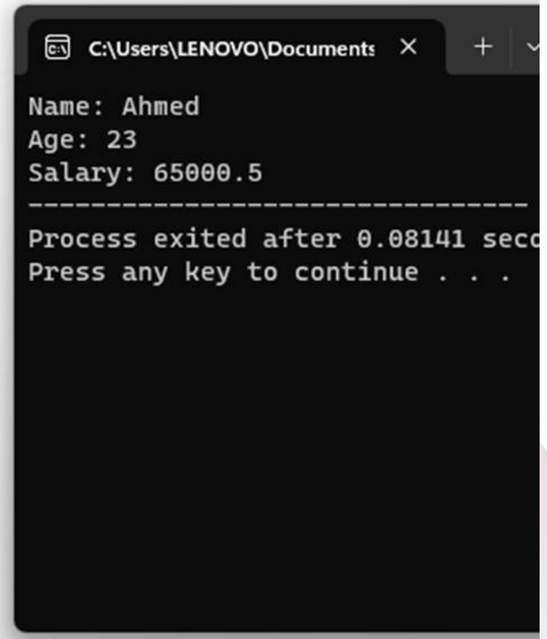
An unnamed structure, also known as an anonymous structure, doesn't have a specific name associated with it.

structure.cpp

```cpp
#include <iostream>
using namespace std;


int main(){
    struct{
    string name;
    int age;
    float salary;} p1;

    p1.name = "Ahmed";
    p1.age = 23;
    p1.salary = 65000.50;
    cout << "Name: " << p1.name << endl;
    cout <<"Age: " << p1.age << endl;
    cout << "Salary: " << p1.salary;
    return 0;
}
```



Structure of Diverse Data:

“Diverse data” means different types of information grouped inside a structure.

A structure can store:

- int
- float
- char
- string
- double
- bool

All inside one structure.

Example:

```
struct Employee {
    int id;
    string name;
    float salary;
    char grade;
};
```

- id → integer
- name → string
- salary → float
- grade → character

Structures as Function Arguments:

We can send a structure to a function just like a simple variable.

There are two ways:

A) Pass Structure by Value

- A copy of structure is passed
- Original structure does NOT change

Example:

```
void display(Student s) {
    cout << s.name;
}
```

CALL:

- display(st1);

Structures as Function Arguments:

B) Pass Structure by Reference

- Address of structure is passed
- Original structure can be modified

Example:

```
void updateMarks(Student &s) {
    s.marks = 90;
}
```

CALL:

```
updateMarks(st1);
```

End Of Class